HEEYEON SUN

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Education

2023 – now **Ph.D. Candidate** in Geosciences (GPA: 4.0/4.0). Dept. of Earth, Geographic, and Climate Sciences, University of Massachusetts Amherst, Amherst, MA, USA. Dissertation: *Multi-Proxy Reconstruction of Holocene Climate Variability in Arid Central Asia using Lacustrine Organic Biomarkers* (degree expected Summer 2027). Advisor: Dr. Isla S. Castañeda

2021 – 2023 **M.S. in Geology** (GPA: 3.87/4.0). Dept. of Geological and Atmospheric Sciences, Iowa State University Ames, IA, USA. Thesis: *Using Clam-based Stable Isotopes and Growth Histories to Reconstruct Oceanographic Variability in the Southern Mid-Atlantic Bight over the last 185 years.* Advisor: Dr. Alan D. Wanamaker.

2013 – 2019 **B.S. in Geological Sciences**, Summa cum laude (GPA: 4.178/4.5). Dept. of Geological and Environmental Sciences, Chungnam National University, Daejeon, South Korea.

2018 Korea Meteorological Administration, Jul 2018 – Aug 2018. Awarded **Certificate of Completion of International Meteorological Professional Training Course**. Seoul, South Korea

2017 Study Abroad Program (GPA: 4.0/4.0). Jan 2017 – Dec 2017. Dept. of Geology, Eastern Illinois University, Charleston, IL, USA.

MAJOR RESEARCH INTERESTS: Organic geochemistry, Paleoclimate, Paleolimnology, Paleoceanography, Stable isotope geochemistry, Paleoclimate proxy development, Holocene climate variability, Hydroclimate variability

Research & Professional Experience

2023 – 2025 **Graduate Research Assistant**, Biogeochemistry laboratory UMass Amherst

- Investigate Holocene climate variability in Lake Issyk-Kul using organic geochemical proxies.
- Analyze GDGTs, alkenones, and leaf-wax biomarkers to reconstruct past temperature and hydroclimate.
- Perform biomarker extraction (ASE) and instrumental analyses (HPLC-MS, GC-MS, GC-FID, GC-irMS).
- Develop age models and conduct statistical analyses and visualization using Python.
- Present findings at international conferences and prepare manuscripts for publication.

2024 – 2026 **Graduate Teaching Assistant**, Univ. of Massachusetts Amherst

Fall 2026 GEOLOGY 445: Sedimentology

- Developed lab materials and graded student assignments.

Fall 2024, 2025, 2026 ENVIRSCI 101: Intro to Environmental Science – led discussions

- Led weekly discussion sections, developed teaching materials, and graded assignments.

Spring 2026 GEOLOGY 105: Dynamic Earth

- Held office hours and provided academic support to students.

Fall 2025, Spring 2026 GEOGRAPH 110: Introduction to Climate Science

- Assisted with lecture materials and led exam review sessions.

2021 – 2023 **Graduate Teaching Assistant**, Iowa State University

Spring 2022, 2023 GEOLOGY 100: How the Earth Works

- Developed lecture materials and graded student assignments.

Spring 2022 ENSCI 382: Analysis of Environmental Systems II

- Assisted lab sections and graded student assignments.

Fall 2021 GEOLOGY 201: Geology for Engineers and Environmental Scientists

- Led weekly lab sections, developed teaching materials and graded assignments.

2021 – 2023 **Graduate Research Assistant**, Stable Isotope Paleo Environments Research Group, Iowa State University

- Reconstructed climate variability in the mid-Atlantic region using stable oxygen isotopes of *Arctica Islandica* shells.
- Conducted high-resolution shell sampling using micromilling techniques for stable isotope analysis.
- Developed shell chronologies to investigate temporal variations in environmental conditions.

2020 – 2021 **Research Assistant**, National Science Museum, Daejeon, South Korea.

- Developed and maintained a database of fossil and rock specimens.
- Identified, classified, and cataloged geological specimens.
- Photographed specimens and documented associated geological information.

2020 **Research Assistant**, Korea Institute of Geoscience and Mineral Resources, Daejeon, South Korea

- Prepared sediment samples for radiocarbon dating.
- Conducted grain-size analysis of sediment samples to characterize sediment properties.

2014 – 2019 **Research Assistant**. CNU Rock Magnetism Lab, Chungnam National University, Daejeon, South Korea.

- Analyzed geological and rock magnetic data to investigate past volcanic activity and

climate variability.

- Presented research findings at national and international conferences.

Publications

Sun, H., Ricketts, D. R., Salacup, J. M., Johnson, T. C., & Castañeda, I. S., (*in prep*). Shifting Holocene moisture sources to Arid Central Asia revealed by compound-specific hydrogen isotopes of plant waxes from Issyk-Kul, Manuscript in preparation for submission to *Geology*. December 2026 expected submission.

Sun, H., Ricketts, D. R., Salacup, J. M., Johnson, T. C., & Castañeda, I. S., The influence of salinity on glycerol dialkyl glycerol tetraether (GDGT) distributions in a large lake (Issyk-Kul) and implications for paleotemperature reconstruction, submitted to *Organic Geochemistry*, October 2026.

Swanner, E. D., Harding, C., Akam, S., Lascu, I., Ledesma, G., Poudel, P., **Sun, H.**, Duncanson, S., Bandy, K., Branham, A.,Bryant-Tapper L., Conwell, T., Jamison, O., & Netz, L., Thermal stratification and meromixis in four dilute temperate zone lakes, *Biogeosciences*, 21, 1549-1562, (2024).

Conference Presentations

13. **Sun, H.**, Ricketts, D. R., Salacup, J. M., Johnson, T. C., & Castañeda, I. S., Shifting moisture sources to Arid Central Asia during the Holocene: a multi-proxy organic geochemical study of Lake Issyk-Kul, Oral presentation, AGU26 Annual Meeting, San Francisco, USA, (2026) (upcoming)
12. **Sun, H.**, Ricketts, D. R., Salacup, J. M., Johnson, T. C., & Castañeda, I. S., Divergent GDGT Proxy Responses Reveal Holocene Climate and Environmental Changes in Lake Issyk-Kul, Central Asia, Poster presentation, Gordon Research Conference on Organic Geochemistry, Holderness, USA, (2026)
11. Remaili, S., Whitney, N., Williams, B., Ummenhofer, C., Biastoch, A., Martin, T., Wanamaker, A. D., Thatcher, D. L., Berg, J., Hauser, A., **Sun, H.**, Mette, M., Walton, A., Powell, E., Mann, R., Hofmann, E. E., Klinck, M. J., Explaining Variability in Shell-Based Oxygen Isotope Ratios in the Western North Atlantic Using a Proxy System Model, AGU25 Annual Meeting, New Orleans, USA, (2025)
10. **Sun, H.**, Castañeda, I. S., Salacup, J. M., & Ricketts, R. D., Spatial-temporal Patterns of the Holocene Climate over Arid Central Asia: Insights from the Lacustrine Records of Eastern Lake Issyk-Kul, Oral presentation, GSA Connects 2025, San Antonio, USA, (2025)
9. **Sun, H.**, Castañeda, I. S., Salacup, J. M., & Ricketts, R. D., Paleoclimate Variability of Lake Issyk-Kul from 14 ka to the Present Based on Organic Biomarkers. Oral presentation, AGU24 Annual Meeting, Washington D.C., USA, (2024)
8. Castañeda, I. S., Daniels, W., Konecky, B., Sae-Lim, J., Salacup, J. M., Schneider, T., **Sun, H.**, Wilk, A., & Zhao, B., Investigating OH-GDGTs and GMGTs in lacustrine sediments and potential relationships to environmental conditions, AGU24 Annual Meeting, Washington D.C., USA,

(2024)

7. Castañeda, I. S., **Sun, H.***, & Salacup, J. M., Temperature and hydroclimate variability in Mid-latitude Arid Central Asia during the past 13,600 years: A multi-proxy investigation of Issyk-Kul (Kyrgyzstan). Oral presentation, GSA Northeastern Section Meeting, Manchester, USA, (2024) *presenting author
6. Thatcher, D. L., Whitney, N. M., Walton, A., **Sun, H.**, Jarosinski, L., Wanamaker, A. D., Lavigne, M., Williams, B., & Stewart, J., Evidence of coherent warming in the northwest Atlantic Ocean from Arctica islandica oxygen isotope records. Poster Presentation, International Sclerochronology Conference, virtual, (2023)
5. **Sun, H.**, Wanamaker, A. D., Thatcher, D. L., Whitney, N. M., Jarosinski, L., Stewart, J. A., Williams, B., & Lavigne, M., Oceanographic Variability in the Rapidly Warming Coastal Mid-Atlantic (USA) over the last 200 years: Insights from Shell-Based Geochemistry and Growth Histories. Poster Presentation, AGU Fall meeting, Chicago, USA, (2022)
4. Thatcher, D. L., **Sun, H.**, Whitney, N. M., Jarosinski, L., Lavigne, M., Williams, B., Stewart, J., & Wanamaker, A. D., Oceanographic changes recorded in Arctica islandica shells from the Mid-Atlantic Bight. Poster Presentation, Virtual International Sclerochronology Conference, (2022)
3. **Sun, H.**, Thatcher, D. L., Whitney, N., Jarosinski, L., Lavigne, M., Williams, B., Stewart, J., & Wanamaker, A., Developing multi-centennial shell-based records from the Mid-Atlantic region. Oral presentation (given virtually), PAGES 6th Open Science Meeting, Agadir, Morocco, OSM18, (2022)
2. **Sun, H.** & Yu, Y., Where do Volcanoes Occur at Destructive Plate Boundaries?. Oral Presentation, Fall Joint Conference of the Geological Sciences, Jeongsun, South Korea, pp. 203, (2014)
1. **Sun, H.** & Yu, Y., Rainfall Dilution: The Solution to Anthropogenic Pollution?. Poster Presentation, Asia Oceania Geosciences Society (AOGS), Sapporo, Japan, SE04-027, (2014)

Awards & Research Funding

2025 **Kerry Kelts Research Award** ($1,500), Geological Society of America (GSA) Limnogeology Division, Boulder, CO, USA. Awarded to one student per year.
Holocene Climate Dynamics in Western Issyk-Kul: Insights from Organic Geochemical Proxies

2024 **GSA Graduate Student Research Grant** ($2,500), The Geological Society of America, Boulder, CO, USA.
Holocene climate variability in Mid-latitude Arid Central Asia: A multi-proxy analysis of organic biomarkers from Lake Issyk-Kul

2023 **Continental Scientific Drilling Facility (CSD) Visiting Graduate Student Program Grant** ($1,000), University of Minnesota, Minneapolis, MN, USA

- Funds supported subsampling and imaging of Issyk-Kul piston and multi-cores for dissertation research.

Other Awards

2026 **Conference Travel Support** ($1,385), Gordon Research Conference on Organic Geochemistry, Holderness, NH, USA

2026 **Leo M. Hall Memorial Fund** ($1,000), Department of Earth, Geographic, and Climate Sciences, University of Massachusetts-Amherst, Amherst, MA, USA
- Award given to support conference presentation.

2024 **Elinor Fierman Memorial Fund** ($700), Department of Earth, Geographic, and Climate Sciences, University of Massachusetts-Amherst, Amherst, MA, USA
- Award given to support taking online coursework in quantitative methods.

2019 **College of Natural Sciences Dean's Award**, Chungnam National University, Daejeon, South Korea

2017 **International Student Scholarship** for excellence in academic study and writing, Eastern Illinois University, Charleston, IL, USA

2017 **College of Sciences Dean's List** in recognition of high academic performance, Eastern Illinois University, Charleston, IL, USA

2013 – 2017 **Scholarships for academic excellence**, Chungnam National University, Daejeon, South Korea

2016 **Global Talent Scholarship for exchange studentship**, Chungnam National University, Daejeon, South Korea

2014 **Scholarship for Excellence Incentive**, Chungnam National University, Daejeon, South Korea

Service and Leadership

2024 – 2025 **Vice President,** Korean Graduate Student Association, UMass Amherst.
- Supported Korean graduate students in adjusting to academic life and promoted Korean cultural awareness in the university community.
- Organized welcome events for incoming graduate students and a career development workshop supported by the Consulate General of the Republic of Korea in Boston.

2024 – 2025 **Undergraduate Mentoring**, UMass Amherst.
- Mentored honors undergraduate student Katherine Eddins in identifying and quantifying primary productivity biomarkers (sterols and long-chain *n*-alkyl diols) in Issyk-Kul sediments.

2025 **Volunteer**, *Eureka!* Summer 2025 Workshop, UMass Amherst.

- Organized and facilitated a hands-on learning station titled "Lakes in Hot Water".
- Developed worksheet questions for student activities.
- Guided students in collecting water samples from Puffers Pond and completing worksheets.

Skills

Laboratory Techniques

- High-performance liquid chromatography-mass spectrometry (HPLC-MS)
- Gas chromatography flame ionization detector (GC-FID)
- Gas chromatography-mass spectrometry (GC-MS)
- GC-isotope ratio monitoring mass spectrometry (GC-irMS)
- The New Wave-Merchantek Micromill
- Thermo Delta V IRMS – GasBench II

Computer Techniques

Python Jupyter ▪ RStudio ▪ MatLab ▪ Adobe Illustrator ▪ LaTeX ▪ HTML ▪ ArcGIS Pro ▪ Adobe Photoshop ▪ Geochemist's Workbench ▪ CooRecorder ▪ CDendro

Professional Memberships

2022 – 2026 American Geophysical Union

2023 – 2026 Geological Society of America

2026 – now Association for Women Geoscientists

2026 – now Asian Americans and Pacific Islanders in Geosciences